CHAPTER

10

Matrices, Retainers and Wedges

Human teeth are designed in such a way that the individual tooth contributes to their own support and collectively support the stomatognathic system. Each tooth is attached in the alveolar bone socket with fine periodontal fibers. These fibers act as cushion and help uniformly dissipating the masticatory forces towards bony tissues. The dissipation of forces is uniform when the teeth are in proper contact and work in unison.

In case the contact is lost, the underlying periodontal fibers and the alveolar bone may not withstand the force, leading to periodontal problems. Lack of stimulation and food impaction, etc. further propagate such problem.

A clear understanding of the interproximal relationship will help the clinician to preserve these structures in a much better manner. To achieve an ideal contact, a clinician should have adequate knowledge of the ideal tooth forms. For example:

- A perfect triangular interproximal space between two adjoining teeth. This space gradually widens out to the labial/buccal and lingual surfaces to form the embrasures.
- Interproximal embrasures are extending on all the four sides, i.e. occlusal, gingival, buccal/labial and lingual having definite shape around each contact area.
- The base of the triangular-shaped interproximal space is located at the alveolar border, while the apex is at the point of contact. Reverse is true for other embrasures, the apex is always at the contact and the base towards the outer surface.
- Anterior teeth usually exhibit marble contacts with less pronounced embrasures.

These ideal conditions are frequently marred by the stresses incident to time, wear, local irritants, configuration disturbances, and imperfectly performed dental operations. The most important function of proximal contact is the protection of the interdental papillae. On anterior teeth, where the papillae form cone-like projections, properly placed point contacts are necessary. A broader buccolingual contact area is required on bicuspids because the crests of the papillae broaden out in this region. Similarly, moving distally, the wider contact area is required on molars because they have the wider interproximal papillae. Improper configuration of the proximal area may lead to:

- Displacement of teeth buccally, lingually, mesially or distally.
- Exert a lifting force on the tooth when placed too high occlusally.
- Disturb the axial relationship of the teeth, resulting in trauma.
- Rotation of the teeth.
- Injury to the investing structures by excessively opening or closing the contact and interproximal embrasures.
- Disturbing the coordination of the inclined planes and cusps, causing defective occlusal contacts.
- Vertical or horizontal food impaction.

TOOTH SEPARATION

Tooth separation implies separating contact of two adjacent teeth, which can be helpful in following situations:

- Examination of interproximal spaces
- Fine preparation of proximal cavities
- Polishing of proximal restorations
- Removal of foreign bodies, such as fruit seeds, fragments of toothpicks, or bone sequestrums , etc.

Two methods, generally employed for accomplishing separation, are:

A. Slow separation
B. Rapid or immediate separation

A. Slow Separation

In this method, the teeth are slowly and gradually forced apart inserting certain materials between them. The advantage of slow separation is that the repositioning occurs physiologically without injuring

periodontal ligament fibers. The disadvantage of this method is that the procedure is time consuming.

Materials used for slow separation are gutta-percha, orthodontic wire, wood or rubber. Gutta-percha may be used in case of adjoining proximal cavity of posterior teeth. Soften the gutta-percha with heat and pack into the cavity overfilling at the proximal side. The material is kept in position for a week and can be renewed, if necessary, until separation is accomplished.

Copper wire, used for orthodontic purpose, can also be passed beneath the contact. The two ends are brought occlusally, twisted together, trimmed and tucked inwards to avoid catching the soft tissues. The wire can accomplish separation in 48 hours.

B. Rapid or Immediate Separation

The rapid separator should be applied carefully to produce desired results. Such a method is more advantageous over slow separation method. Though, the method is quick and useful in clinical conditions, yet it may rupture the periodontal ligament fibers and rapid separation may induce pain at the site. The rapid or immediate separation is achieved following two principles, viz. wedge principle and traction principle.

a. *Separation by Wedge Principle*

The separation by 'wedge principle' is accomplished by insertion of a pointed wedge-shaped device between teeth in order to create space at the contact area. The more the wedge moves facially or lingually, the greater will be the separation. This separation is brought about by mechanical device (Elliot separator) along with wedges.

i. *Elliot separator:* Many a times, the separation is required for short duration. The Elliot separator, which work on wedge principle, is effective in such conditions (Fig. 10.1). During its application, care must be exercised to prevent slipping. This type of separation is useful in examining a proximal surface or for final polishing of the restoration.

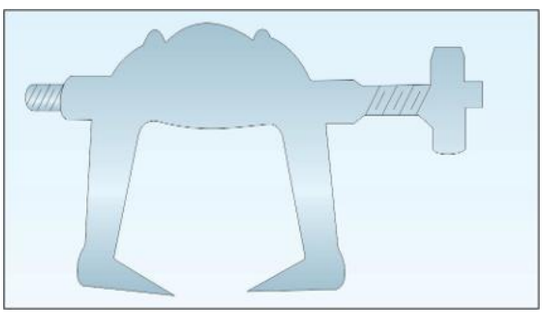


Fig. 10.1: Elliot separator

ii. *Wedges leading to separation by wedging principle include:*
- Wooden wedges
- Metal wedges
- Silver wedges
- Celluloid or plastic wedges
- Medicated wood wedges

b. *Separation by Traction Principle*

The separation by 'traction principle' is accomplished with mechanical devices, which engage the proximal surfaces of the teeth to be separated by means of holding arms. These are mechanically moved apart, creating separation between the teeth.

The separators, which work on traction principle, are:

i. Non-interfering true separator
ii. Ferrier double bow separator
iii. Ivory adjustable separator
iv. Perry separator
v. Woodward separator
vi. Parr's universal separator
vii. Dentatus-Nystrom separator

i. ***Non-interfering true separator:*** This device is indicated when continuous stabilized separation is required during the dental operation. Its advantage is that the separation can be increased or decreased after stabilization, and the device is non-interfering (Fig. 10.2).

ii. ***Ferrier double-bow separator:*** With this device, the separation is stabilized throughout the operation. Its advantage is that the separation is shared by the contacting teeth, and not at the expense of one tooth, as with the non-interfering separator (Fig. 10.3).

The separators from (iii) to (vii) are not used in routine.

MATRICES

The word matrix is derived from the Latin word 'Mater', which means 'Mother'; introduced by Dr. Louis Jack (1871).

The matrix is a device used to contour a restoration to simulate the tooth morphology, which it is

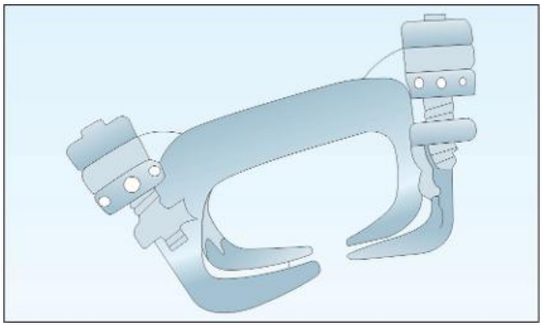


Fig. 10.2: Non-interfering true separator

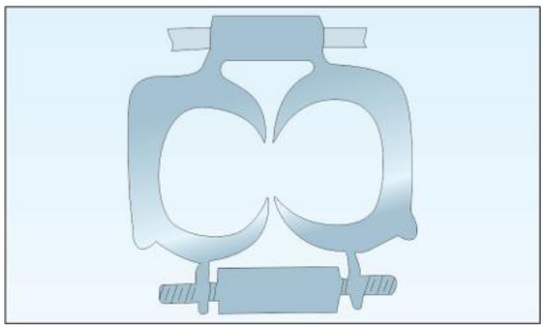


Fig. 10.3: Ferrier double bow separator

replacing. *Matricing is the procedure, whereby a temporary wall is created opposite to axial walls and surrounding areas of tooth structure that were lost during preparation.*

Requirements

The matrix should create the exact three-dimensional contour (including the contact area) of the future restoration. It should be immobile while the material sets; also should not react with the material. It should be easily removable after hardening of restorative material without compromising the created contact, contour and characteristics of the restoration.

Requirements of matrices

- Should be inserted easily and be sufficiently rigid.
- Should retain its contour, which can easily be transferred to the restoration.
- Should not adhere to or react with the restorative material.
- Should resist the condensation pressure.

Matrix Materials

The materials used as matrices include stainless steel, cellulose acetate (cellophane), cellulose nitrate (celluloid) and polymer materials. Matrix system is mostly formed of two parts: A band, which is made of metal, polymeric material or celluloid (Fig. 10.4A, B) and a retainer. Matrices are commonly supplied as strips of different dimensions. They may be 0.001–0.0015 inch (0.025 mm or 0.002 mm) thick. The widths of the matrix band, mostly marketed, are 1/4 , 3/8, 5/16 and 1/8 inch.

Objectives

The objectives of matricing are:

- Should act as a temporary wall of resistance during introduction of the restorative material.
- Should provide shape to the restoration.
- Should confine the restoration within acceptable physiological limits.

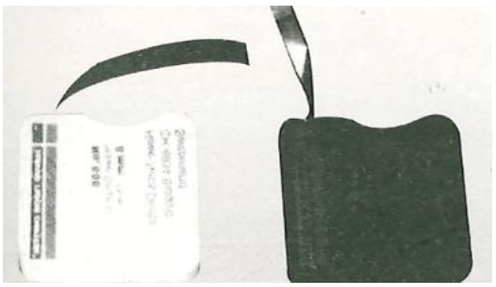


Fig. 10.4A: Stainless steel and plastic matrices



Fig. 10.4B: Matrices

- Help in maintaining the dry operative field thereby preventing contamination of the restoration.
- Assist in isolating the gingiva and rubber dam during introduction of the restorative material.

Classification

Matrices are classified in two ways; one is based on mode of retention and second is based on transparency.

a. Matrices based on mode of retention
   i. Mechanically retained matrices
   ii. Self-retained matrices
b. Matrices based on transparency
   i. Non-transparent matrices
   ii. Transparent matrices

MATRICES FOR CLASS II, MOD AND COMPLEX RESTORATIONS

Early mechanical matrices were Miller's clamp matrix, Woodward's screw matrix and the loop matrix (Fig. 10.5A to C). These matrices were not flexible and their insertion around the tooth was difficult; however, they led to separation of teeth. The major disadvantage was that the contact and contours were lost once they were drawn onto the teeth. This led to the advent of stainless steel matrices (0.002 inch thickness) to be screwed with the help of retainers. The disadvantages of early matrices were eliminated, since the new matrices could be contoured according to the contour of the tooth to be restored.

A few authors are of the opinion that the matrix be held without the use of retainers; only wedges are

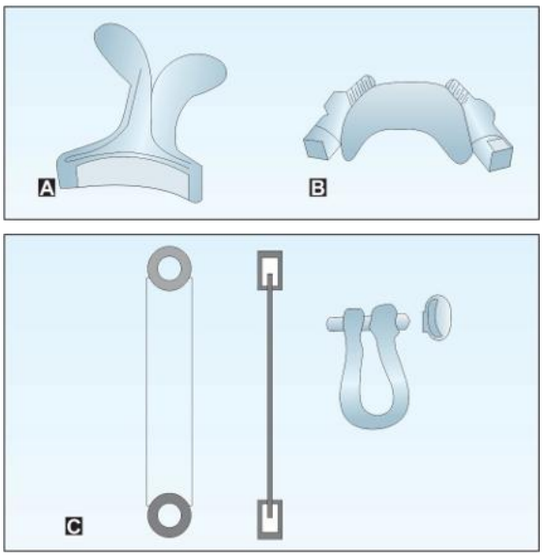


Fig. 10.5: **(A)** Miller's clamp matrix; **(B)** Woodward's screw matrix; **(C)** Loop matrix

sufficient. The rationale is that the retained matrices usually produce straighter proximal areas and matrix held with only wedges produce better contours and contacts.

For MOD preparation and complex restoration, a continuous matrix band is indicated. Such a matrix band may be retained with a mechanical holder and can be ligated. Copper bands (Fig. 10.6A), used for such purposes, can be trimmed with scissors, smoothened and placed onto the teeth (Fig. 10.6B).



Fig. 10.6A: Copper band

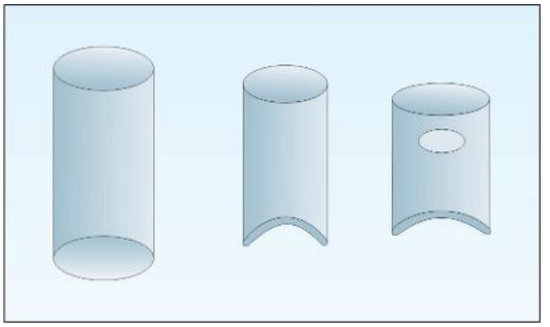


Fig. 10.6B: Copper band matrix (trimmed)

These are mostly used with silver amalgam restorations involving more than two surfaces.

Sectional Matrix System

The sectional matrix system and separation rings (mostly used for Class II composites) are made up of nickel-titanium alloy. The system creates a consistent force to separate teeth and then return to their original shape after use; helping to deliver anatomically contoured restoration. Some of the commercially available systems are:

i. Conventional (Fig. 10.7A) and saddle matrix system (Fig. 10.7B)
ii. Palodent Plus sectional matrix system (Fig. 10.7C)
iii. Composi-Tight 3D sectional matrix system (Fig. 10.7D)
iv. Triodent V3 Ring matrix system (Fig. 10.7E).



Fig 10.7A: Conventional sectional matrix



Fig 10.7B: Saddle matrix system



Fig 10.7C: Palodent Plus sectional matrix system



Fig. 10.7D: Composi-Tight 3D sectional matrix system



Fig. 10.7E: Triodent V3 Ring matrix system

The technique for the placement of Palodent Plus sectional matrix is as follows (Fig. 10.8A to H):

- Place a matrix band that closely approximates the occlusogingival height of the tooth. The band



Fig. 10.8A: Palodent retainer



Fig. 10.8B: Palodent holder



Fig. 10.8C: Palodent system for permanent teeth

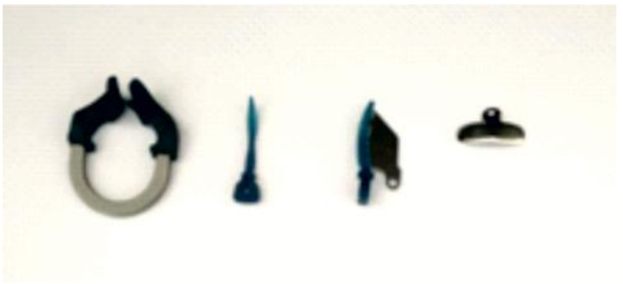


Fig. 10.8D: Palodent system for deciduous teeth



Fig. 10.8E: Palodent ring placed



Fig. 10.8F: Palodent band and wedges



Fig. 10.8G: Wedge placed from both the sides



Fig. 10.8H: Double retainer for MOD cavity

should be oriented with concave edge towards the occlusal margin of the tooth.

- Insert wedge.
- Apply the 3D-ring retainer. Hold the ring retainer with the ring placement forcep and place it over the wedge.
- Burnish the band in the desired contact area against the adjacent tooth and make sure there is no spring-back of the band.
- Restore the cavity as desired.
- Remove the ring, wedge and the band (removal of the ring and band may need forceps).
- For MOD preparations, two rings can be placed (Fig. 10.8H).

MATRICES FOR CLASS III RESTORATIONS

The matrices used for Class III restorations are usually cellophane or mylar strips.

The strip is burnished over the end of a steel instrument; for example, handle of a tweezer, to produce a 'belly' in the strip. This will allow for a curvature, which, if properly contoured and designed, will reproduce the natural proximal contour of the tooth (Fig. 10.9A and B).

In distal surface of canine, since the fixation of retainer is difficult, a metal band is molded into 'S' shape and stabilized using wedges and/or impression compound (Fig. 10.10).

MATRICES FOR CLASS IV RESTORATIONS

A suitable strip is folded and molded into L shape. Longer side of the strip is cut according to length of the tooth and the short side is cut according to width of the tooth. The strip is adapted to the tooth and wedge placed. It is important that the angle formed by the fold of the strip approximates the normal corner of the tooth; the lingual surface is supported by the forefinger of the left hand. The cavity is then

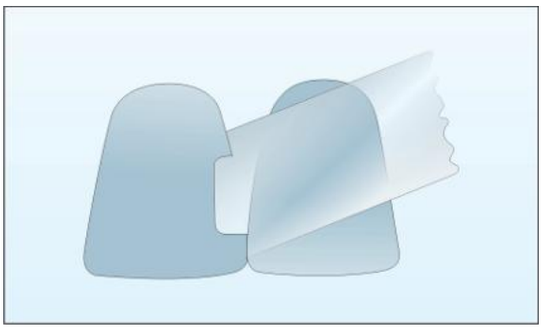


Fig. 10.9A: Matrix for Class III restoration



Fig. 10.9B: Matrix strip applied around left maxillary canine (on model)

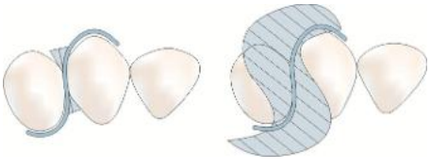


Fig. 10.10: 'S' shaped matrix for distal surface of canine

filled to slight excess, and one end of the strip is brought across the proximal surface of the filled tooth.

Prefabricated matrices are also available, or can be fabricated to suit the required restoration. Commonly available matrices are:

i. ***Aluminum foil incisal corner matrix:*** These are 'stock' metallic matrices, shaped according to the proximoincisal area of anterior teeth (Fig. 10.11). This type of matrix cannot be used for light-cured resin material.

ii. ***Transparent crown form matrices:*** These are 'stock' plastic crowns, which can be adapted to tooth anatomy (Fig. 10.12). In bilateral Class IV preparations, use the entire crown form; but in a unilateral preparation, cut the plastic crown incisogingivally into two halves and use only the side corresponding to the location of the preparation.

iii. ***Anatomic matrix:*** Before tooth preparation, a study cast for the affected tooth along with adjacent teeth is prepared.

The defective areas are restored on the study model by plaster of acrylic resin. A plastic template is made over the restored tooth/teeth on the model. The template is trimmed gingivally to fit the tooth. It should seat on at least one unprepared tooth on each side. The prepared matrix should be vented by perforating the corners. The restorative material is filled in the preparation, and the matrix filled with the material is pressed over the partially filled tooth. The restoration is cured while the matrix is in place.

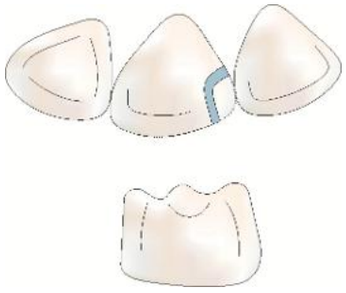


Fig. 10.11: Matrices for Class IV cavity

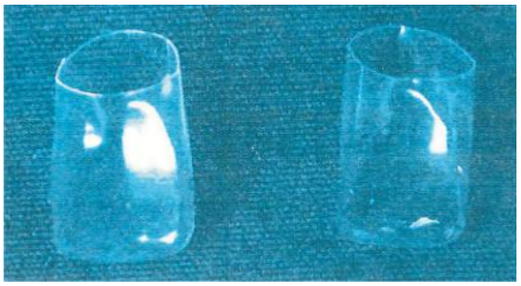


Fig. 10.12: Transparent celluloid crowns

The matrix is removed or cut and the final restoration is polished.

MATRICES FOR CLASS V RESTORATIONS

The matrices for Class V restorations are available as prefabricated or can be fabricated outside the oral cavity. The commonly available matrices for Class V restoration are:

i. ***Prefabricated plastic matrices:*** These are available in different sizes. A handle is provided to hold the matrix in place till the material sets (Fig. 10.13). The clinical application of this matrix is shown in Fig. 10.14.

ii. ***Aluminum/copper collars:*** Aluminum/copper collars are used for materials, which do not require light curing. These are preshaped according to the gingival third of the buccal and lingual surfaces. They can be adjusted to cover 1.0–2.0 mm of the tooth surface circumferential to the cavity margins. Fill the cavity with restorative material and apply the adjusted collar onto the tooth.



Fig. 10.13: Prefabricated plastic cervical matrices



Fig. 10.14: Adaptation of cervical matrix to a Class V cavity

iii. *Anatomic matrix:* Anatomic matrix can be fabricated as for Class IV cavities. The template is cut mesiodistally and trimmed gingivally. The prepared matrix is used to apply pressure while the material is being cured.

MATRIX RETAINERS

Matrix retainers are gadgets used to retain the matrix bands in position. Some matrices do not need mechanical devices to hold them in position. Some matrices may require simple retainers like wires, silk thread, dental floss and impression compound. Some matrices do need mechanical retainers.

a. Mechanical Retainers

The types of mechanical retainers used are as follows:

i. *Nystrom's retainer:* The shape ensures a 20°–30° inclination of the retainer in relation to the band (narrowest part faces the gingiva). The retainer is no longer used.

ii. *Ivory matrix retainer No. 1 and 8:* Ivory Matrix Retainer no. 8 provides bands for encircling entire crown of the tooth (suitable for Class II cavities and for mesio-occluso-distal cavities). Ivory matrix no.1 encircle one proximal wall (suitable for Class II cavities) (Fig. 10.15A, B). The application of these retainers is shown in Fig. 10.16A, B.

iii. *Steele's siqveland self-adjusting matrix retainer:* The matrix retainer forms two diameters of the band at the same time; larger diameter for the occlusal end and smaller for the gingival end. Anatomic adaptation is possible without wedges. The band follows the tooth contour without impinging on the gingival tissue (Fig. 10.17).

iv. *Tofflemire universal matrix band retainer:* Tofflemire (Figs 10.15C and 10.18) is stable and

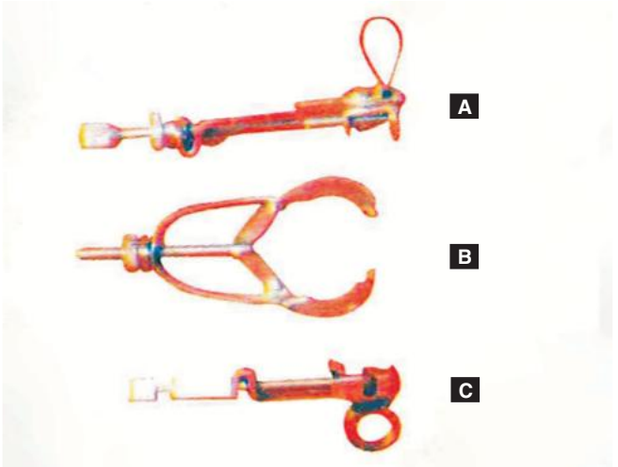


Fig. 10.15: Retainers from top to bottom: **(A)** Ivory No. 8, **(B)** Ivory No. 1, **(C)** Tofflemire



Fig. 10.16A: Application of Ivory No. 8 retainer and band



Fig. 10.16B: Application of Ivory No. 1 retainer

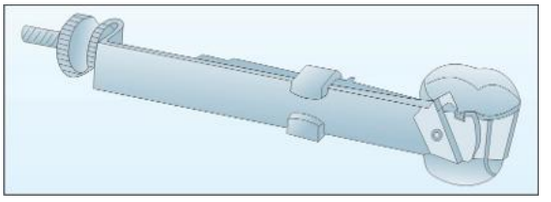


Fig. 10.17: Siqveland matrix retainer

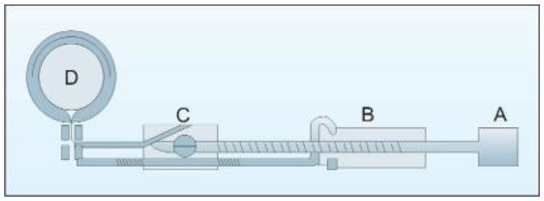


Fig. 10.18: Tofflemire matrix retainer

sturdy; permits easy removal of the holder from the band.

Operating instructions are as follows:

- Turn the knurled nut (B) to the right until the diagonal slot (C) is about ¼ inch from the inner end of the retainer.
- Hold the knurled nut (B) from rotating while the knurled nut (A) (at the end of the spindle) is turned a like number of turns in the opposite direction (left), until the point of the spindle clears the diagonal slot channel for the reception of the free ends of the matrix band.
- Insert the 'occlusal edge' of whichever type of matrix band (D) is decided in the diagonal slot (C). The loop is placed in such a manner that the metal arch of the guide channel serves as an occlusal 'stop'. Various sizes of bands are available, as per requirement of the case.

Fitting band to the teeth

- Guide the band gently over the tooth, using the retainer as a carrier. Should the loop be too small to pass over the contour of the tooth, turn the knurled nut (B) a turn or two to the left, and the loop will be enlarged. Conversely, the loop is decreased in size, or tightened around the tooth, by turning the knurled nut (B) to the right.
- After completing the restoration and the preliminary finishing, the retainer is removed from the band without disturbing the band at this time. This is accomplished by holding the knurled nut (B) steady while the knurled nut (A) is turned a few turns to the left, thereby permitting the retainer to slip off the band (Fig. 10.19A–H).
- The band is removed carefully from the contact point. Support the occlusal surface of the freshly condensed restoration, then gently ease the band using a lateral rotation motion rather than an occlusal traction motion. The application of Tofflemire retainer is shown in Fig. 10.20.

b. Automatrix

Automatrix, also known as roll-in band matrix, is a disposable system where band and retainer are constructed as one unit. Bands of different lengths, widths and thicknesses are available (Fig. 10.21) (length: 4.7 to 7.9 mm and thickness: 0.38 to 0.05 mm). The matrix along with the tightening device is shown in Fig. 10.22. The instability of this system renders it less suitable for routine use. The automatrix system is primarily useful in patients who cannot tolerate retainers and in patients with partly erupted teeth, where the height of the tooth provides insufficient support for the retainers.

WEDGES

Wedges are the third component of the matrix system. Wedging serves the following purposes:

- Prevents surplus amalgam being forced into the gingival crevice.
- Assists in contouring the cervical part of the proximal surface.
- Separates the teeth to compensate for the thickness of the matrix band, such that proximal contact is re-established when the band is removed.
- Stabilizes the matrix.

Classification

a. ***Wooden or plastic wedges:*** Wedges are made of wood or plastic. Wooden wedges are preferred because:
   - They are easy to trim. They adapt well to the tooth surface.
   - They remain stable during condensation.

b. ***Preformed or custom made wedges:*** Preformed wedges are commercially available in different sizes and are preferred because of their convenience of use.
   The requisite size of custom-made wedges may be fabricated from tooth picks.

c. ***Triangular (anatomic) or round wedges:*** In general, a wedge must be triangular or round in cross-section (Fig. 10.23). The width of the base should be slightly larger than the cervical embrasure. If the wedge is not high enough, only point contact between the wedge and the band is achieved. This may lead to poor contour or displacement of the wedge during condensation. A uniform tapering of the wedge is needed in order to render sufficient and even contact throughout the proximal embrasure.

d. ***Opaque or transparent:*** Transparent (resin) wedges can be used during placement of composite resin restorations as they allow visible curing light to pass through.

Placement of Wedges

In general, the wedge is inserted from the lingual, as this embrasure is normally larger in size. However,

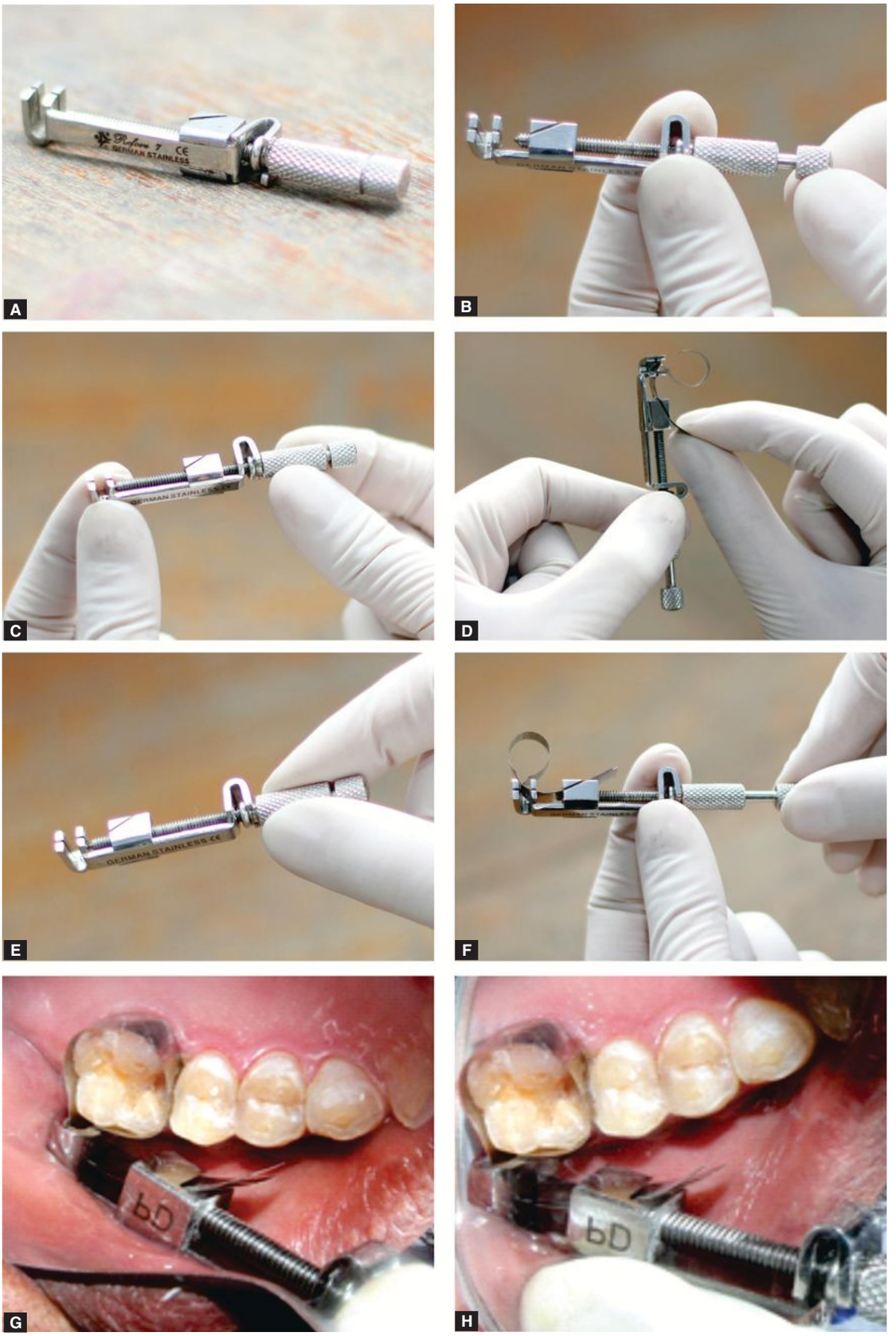


Fig. 10.19: Use of Tofflemire retainer: Steps followed



Fig. 10.20: Application of Tofflemire retainer and band

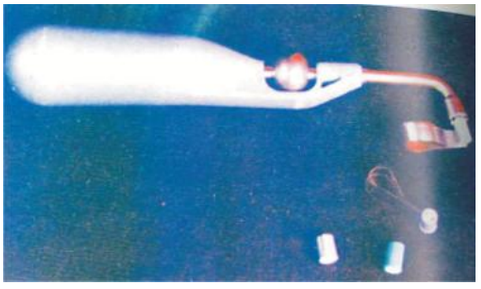


Fig. 10.21: Automatrix system

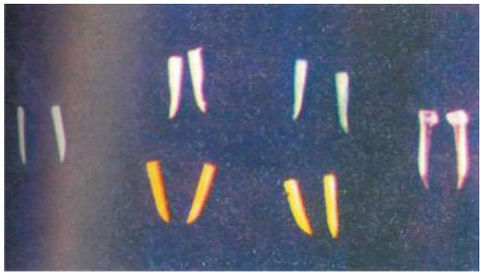


Fig. 10.22: Tightening device for automatrix

since the lingual wedge will interfere with the tongue, it is preferred form the buccal side. In case of maxillary teeth, placement of wedge is preferred from palatal side. During insertion, care should be taken to ensure that the wedge is apically positioned in relation to the gingival cavity wall. The clinician should have an adequate knowledge of the anatomical and functional aspects of contacts and contours so as to reproduce them with different restorative materials (Fig. 10.24A, B).

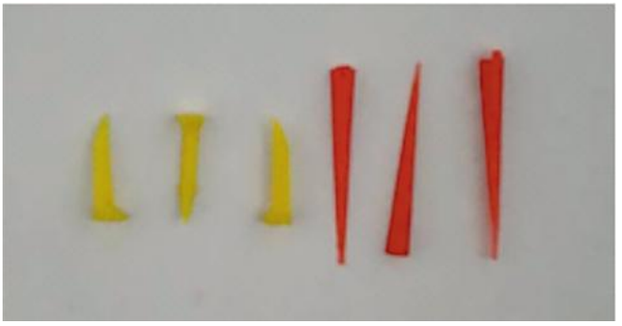


Fig. 10.23: Types of wedges

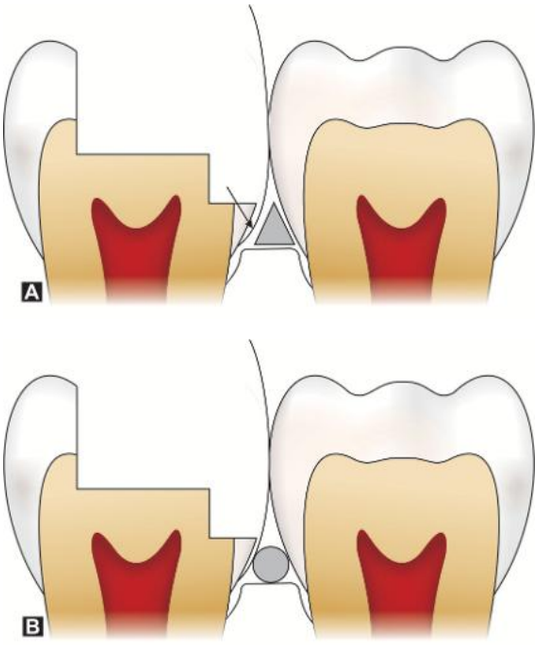


Fig. 10.24: **(A)** As a rule, the triangular wedge does not firmly support the matrix band against the gingival margin in conservative Class II preparations (arrow). **(B)** The round toothpick wedge is preferred with these preparations because its wedging action is nearer the gingival margin

Wedging Systems

1. *'Piggyback' wedging:* If the wedge is significantly apical of the gingival margin, a second, usually smaller wedge may be 'piggybacked' on the first wedge. 'Piggyback' wedging is particularly useful in patients with recession of interproximal area (Fig. 10.25).
2. *Wedge-wedging system:* Occasionally, a concavity may be present on the proximal surface gingivally of the contact and extending as fluting onto the root (e.g. mesial of the maxillary first premolar). A gingival margin located in this area will be similarly concave. To wedge a matrix band tight against such a margin, a second pointed wedge can be inserted between the first wedge and the band by 'wedge

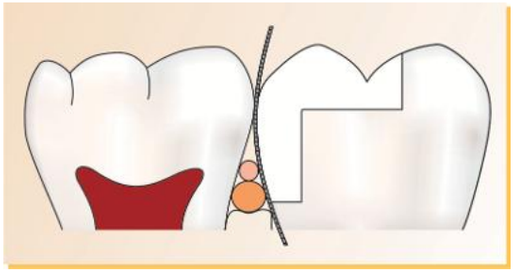


Fig. 10.25: Piggyback wedging: Smaller wedge may be piggybacked on first wedge

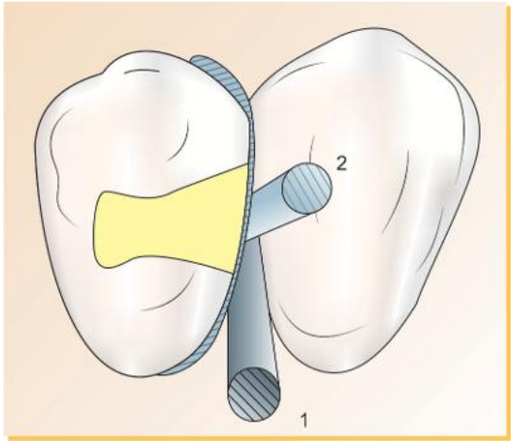


Fig. 10.26: Wedge-wedging system: 1. Conventional wedge; 2. Vertical wedge applied between the matrix band and wedge 1

wedging'. The wedging action between the teeth should provide enough separation to compensate for the thickness of the matrix band (Fig. 10.26).

3. ***Double wedging system:*** In teeth with facio-lingually wide proximal boxes, traditionally double

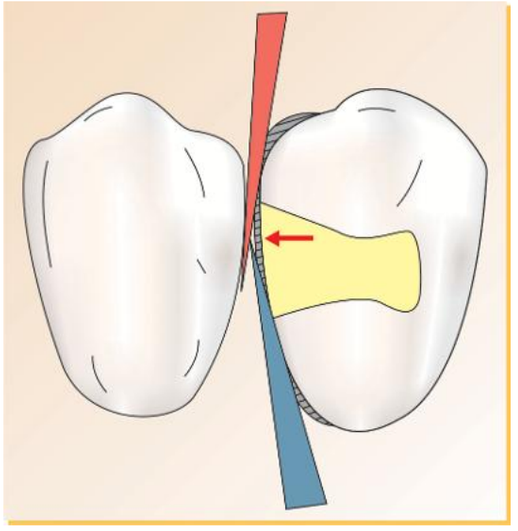


Fig. 10.27: Double wedging system: The arrow indicates small space leading to overhanging of restorative material at that area

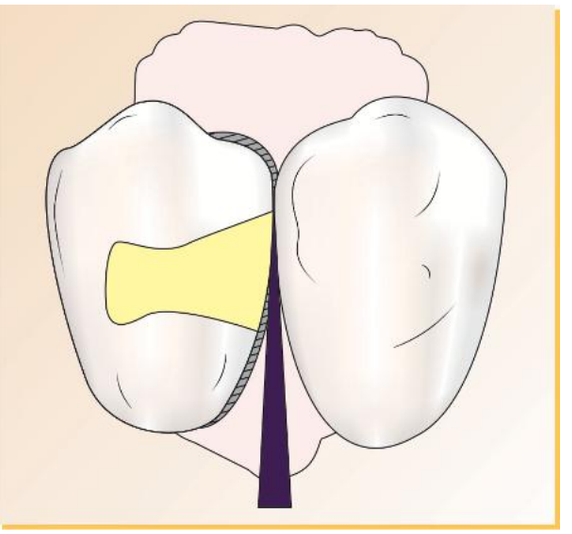


Fig. 10.28: Compound supported matrix band

wedging has been proposed. It refers to insertion of wedges—one from the lingual embrasure and one from the buccal embrasure. However, it has been observed that insertion of wedges from both embrasures leaves some space just below the contact area leading to overhanging of the restorative material at that area (Fig. 10.27). Since, insertion of wedges from buccal embrasure in maxillary posterior teeth is difficult and also to avoid double wedging, the use of compound supported matrix should be preferred (Fig. 10.28).

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